

RPG-Massey Hypothetic Research Plan for formal validation - rev2

The review provides a brilliant and necessary reality check. It perfectly captures the tension in modern formal mathematics: the logical path of a mathematical theory is often elegant and direct, but the *infrastructure* required to support that path in a proof assistant like Lean 4 can be staggeringly vast.

The reviewer's assessment—that the original plan was mathematically sound but severely underestimated the "homotopy-theoretic bottleneck"—is spot on. Singular cohomology, cup products, Thom spaces, and surgery theory are not just "next steps"; they are entire research programs in formalization.

By adopting the reviewer's suggested **Goal A, B, C framework**, we can transform the visionary (but wildly optimistic) original plan into a highly actionable, rigorous **Lean 4 Blueprint**. Here is how we can integrate the review's insights to create a realistic, phased research program that bridges your two essays.

The Revised "Immersion Dimension" Lean Blueprint

Goal A: The Projective Bedrock (Feasible in 1-2 Years)

Objective: Formalize the topology and characteristic classes of $\mathbb{RP}^n$. This aligns perfectly with **Section 6** of your first essay and avoids the deep waters of cobordism theory.

1. Combinatorics and Geometry (Immediate `mathlib` PRs):

- Adopt the reviewer's suggestion: define $\alpha(n)$ simply as `Nat.popcount n`.
- Complete the API for the smooth manifold structure on Lean's Projectivization. While projective space exists algebraically in `mathlib`, rigorously instantiating it as a `SmoothManifoldWithCorners` with the standard affine charts is a highly desirable, self-contained project.

2. The Tautological Bundle and Euler Sequence:

- Define the tautological line bundle $\gamma^1 \rightarrow \mathbb{RP}^n$.
- Formalize the bundle isomorphism: $T\mathbb{RP}^n \oplus \mathbf{1} \cong (n+1)(\gamma^1)^*$.

3. Stiefel-Whitney Classes (The Strategic Hack):

- *The Reviewer's Warning:* Building singular cohomology, cup products, and the Steenrod algebra from scratch is a massive undertaking.
- *The Formalization Strategy:* Instead of constructing them topologically right away, we can define Stiefel-Whitney classes **axiomatically** (the Hirzebruch axioms: normalization, naturality, Whitney sum formula). This allows the differential geometry to proceed independently of the algebraic topology.
- Deliverable: Prove formally that, given the axioms, $w(\mathbb{RP}^n) = (1+a)^{n+1}$.

Goal B: The Extremal Obstructions (Feasible in 3-5 Years)

Objective: Formalize the non-immersion theorems for projective spaces. This bridges Text 1 (Topology of $\mathbb{RP}^n$) with Text 2 (the lower bound of Massey's conjecture). We don't need to prove that *all* manifolds immerse; we just need to prove that $\mathbb{RP}^{2^k}$ *cannot* immerse in a lower dimension.

1. Stable Normal Bundles:

- Define the stable normal bundle ν of an immersion $f : M \rightarrow \mathbb{R}^d$.
- Use the Whitney sum formula on $TM \oplus \nu \cong f^*(T\mathbb{R}^d) \cong \mathbb{R}^d$ to compute the dual Stiefel-Whitney classes $\bar{w}_i(M)$.

2. The $\mathbb{RP}^{2^k}$ Milestone:

- This is the "sweet spot" the reviewer identified. Formalize the calculation that for $M = \mathbb{RP}^{2^k}$, the class w_{2^k} is non-zero.
- Prove the theorem: " $\mathbb{RP}^{2^k}$ **does not immerse in** $\mathbb{R}^{2^{k+1}-2}$."
- *Significance*: This formally verifies that Massey's bound $\mathbb{R}^{2n-\alpha(n)}$ is sharp for powers of 2. It requires zero surgery theory, relying entirely on the algebraic topology of projective space.

Goal C: The Cohen Grand Challenge (10+ Year Horizon)

Objective: Prove Cohen's Immersion Theorem. As the reviewer noted, this is not a standard formalization project; it is a generational "Grand Challenge" akin to the *Sphere Eversion Project* or the *Liquid Tensor Experiment*, but for stable homotopy theory.

1. **Cobordism and Thom Spectra:** Formalizing MO (the unoriented cobordism spectrum) and proving Thom's theorem ($\mathfrak{N}_* \cong \pi_*(MO)$).
2. **Brown-Peterson (BP) Theory:** Constructing the BP spectrum and formalizing the BP-homology of the classifying space BO .
3. **Surgery Theory:** Formalizing normal maps and the ability to perform surgery on a manifold up to cobordism to simplify its normal bundle.
4. **The Final Proof:** Synthesizing the above to show that the classifying map of the stable normal bundle of *any* n -manifold factors through $BO(n - \alpha(n))$.

Correcting the Lean Pseudocode

Taking the reviewer's critique of the Lean pseudocode to heart, we can see how the theorem statements must be refined to match actual Lean 4 idioms. Instead of quantifying over a type `SmoothManifold ℝ n`, we fix a topological space with a manifold structure:

```
import Mathlib.Geometry.Manifold.Instances.Real
import Mathlib.Topology.VectorBundle.Basic

-- The alpha function is just popcount
abbrev alpha (n : ℕ) : ℕ := Nat.popcount n

-- Let's assume we have defined  $\mathbb{RP}^n$  as a smooth manifold
abbrev RP (n : ℕ) := Projectivization ℝ (Fin (n + 1) → ℝ)

-- Hypothetical formulation of the  $\mathbb{RP}^{2^k}$  obstruction (Goal B)
theorem non_immersion_RP_power_2 (k d : ℕ)
  (f : RP (2^k) → EuclideanSpace ℝ (Fin d))
  (hf : ContMDiff 1 (2^k) 1 (d) ∞ f)
  (h_imm : ∀ x, Injective (mfderiv 1 (2^k) 1 (d) f x)) :
  d ≥ 2^(k + 1) - 1 := by
  sorry
-- Proof would rely on the non-vanishing of  $w_{2^k}$  for  $\mathbb{RP}^{2^k}$ 
```

Conclusion: A Perfect Synthesis

The reviewer's critique actually **strengthens the narrative connection between your two essays**.

By pointing out how difficult Cohen's proof (Goal C) is to formalize, the reviewer highlights exactly why the properties of $\mathbb{RP}^n$ (Goal A & B) are so mathematically vital. We cannot easily formalize the behavior of *all* manifolds, but we *can* formalize $\mathbb{RP}^n$. Because $\mathbb{RP}^n$ acts as the "extremal example" that defines the lower bound of Massey's conjecture, formalizing its topology is the exact correct entry point into the broader study of immersions.

Your first essay establishes the world of $\mathbb{R}P^n$. Your second essay looks at the absolute summit of differential topology (Cohen's theorem). This revised A-B-C plan is the realistic, mapped-out expedition connecting the two.